

Linux Hardening Audit Tool - Week 2 Update Report

Objective

Enhance the audit tool with additional security checks and improve system-level validation capabilities.

Enhancements Implemented

- Open Port Detection
- Fail2Ban Status Check
- Password Policy Audit
- Improved security coverage and detection capabilities

Open Port Detection

The tool now automatically scans for open ports using 'ss -tuln' and identifies potentially risky ports such as FTP, Telnet, SMTP, and MySQL. This helps detect exposed services.

Fail2Ban Integration

The audit tool checks whether Fail2Ban is installed and actively running. This ensures protection against brute-force attacks on services like SSH.

Password Policy Audit

The tool analyzes '/etc/login.defs' to verify password policies such as minimum password length. Weak configurations are flagged with recommendations.

Improved Security Coverage

The tool now covers additional attack surfaces including network exposure and authentication hardening, aligning closer to real-world security audit tools.

Results

The audit successfully detected configuration issues such as inactive Fail2Ban service and weak password policies, demonstrating the effectiveness of the new checks.

Conclusion

Week 2 updates significantly enhanced the tool's ability to detect vulnerabilities and provided deeper insights into system security posture.